Note B: baseline and robustness results

Unemployment and prices after a monetary policy surprise

Generated by `baseline/05_report.py`

Specification

Each number is the response of the unemployment gap (percentage points) or the core PCE price *level* (percent) to a monetary policy surprise that raises the 2-year Treasury yield by 1 percentage point in the announcement window, 12, 24 and 36 months later. Responses are traced through a daily block (the 2-year yield and, where included, the market financial conditions index (FCI) of Note A) into a monthly system for the unemployment gap and core PCE inflation. **Main tables use 12 monthly lags, fixed in advance; the appendix shows 3, 6 and 12.** The surprises are the Nakamura–Steinsson policy news series (252 FOMC announcements, January 1996 to July 2026), identified in the announcement window with macro variables held fixed. Macro variables come first within the month. No deterministic trend is included except in Robustness 2c–d. The sample runs to September 2026 unless stated. Intervals are 90% bootstrap intervals from 200 draws (resampling announcements and monthly residuals; the daily block is not re-estimated). A star marks a 90% interval that excludes zero. Without the FCI the daily block contains the 2-year yield alone; with it, the FCI is in the same form (nominal, real or r*-gap) as the 2-year yield.

1 Baseline

Table 1: Baseline: 1994 to the present, all data in, no trend (12 lags)

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Nominal	none	12	392	−0.70*	−0.38*	−0.20	0.08	0.01	−0.12
				[−1.17, −0.28]	[−0.72, −0.08]	[−0.56, 0.09]	[−0.10, 0.26]	[−0.30, 0.34]	[−0.55, 0.38]
	nominal	12	392	−0.02	0.24	0.32*	−0.09	−0.33	−0.57*
				[−0.37, 0.39]	[−0.04, 0.53]	[0.07, 0.55]	[−0.32, 0.12]	[−0.77, 0.03]	[−1.19, −0.09]
Real	none	12	392	−0.54*	−0.28	−0.16	−0.03	−0.10	−0.21
				[−0.92, −0.20]	[−0.58, 0.00]	[−0.50, 0.13]	[−0.20, 0.13]	[−0.40, 0.19]	[−0.63, 0.26]
	real	12	392	0.07	0.25	0.26	−0.18*	−0.38	−0.54*
				[−0.30, 0.43]	[−0.05, 0.48]	[−0.02, 0.47]	[−0.38, 0.00]	[−0.74, 0.03]	[−1.14, 0.00]
r*-gap	none	12	392	−0.28	−0.06	0.04	−0.18	−0.39*	−0.62*
				[−0.77, 0.08]	[−0.54, 0.30]	[−0.41, 0.45]	[−0.34, 0.03]	[−0.68, 0.00]	[−1.11, −0.06]
	r*-gap	12	392	0.31	0.47*	0.41*	−0.36*	−0.71*	−0.97*
				[−0.03, 0.72]	[0.15, 0.75]	[0.09, 0.70]	[−0.54, −0.20]	[−1.11, −0.34]	[−1.64, −0.33]

Nominal, real and r*-gap 2-year yield; the FCI is Note A's index in the matching form in each case. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

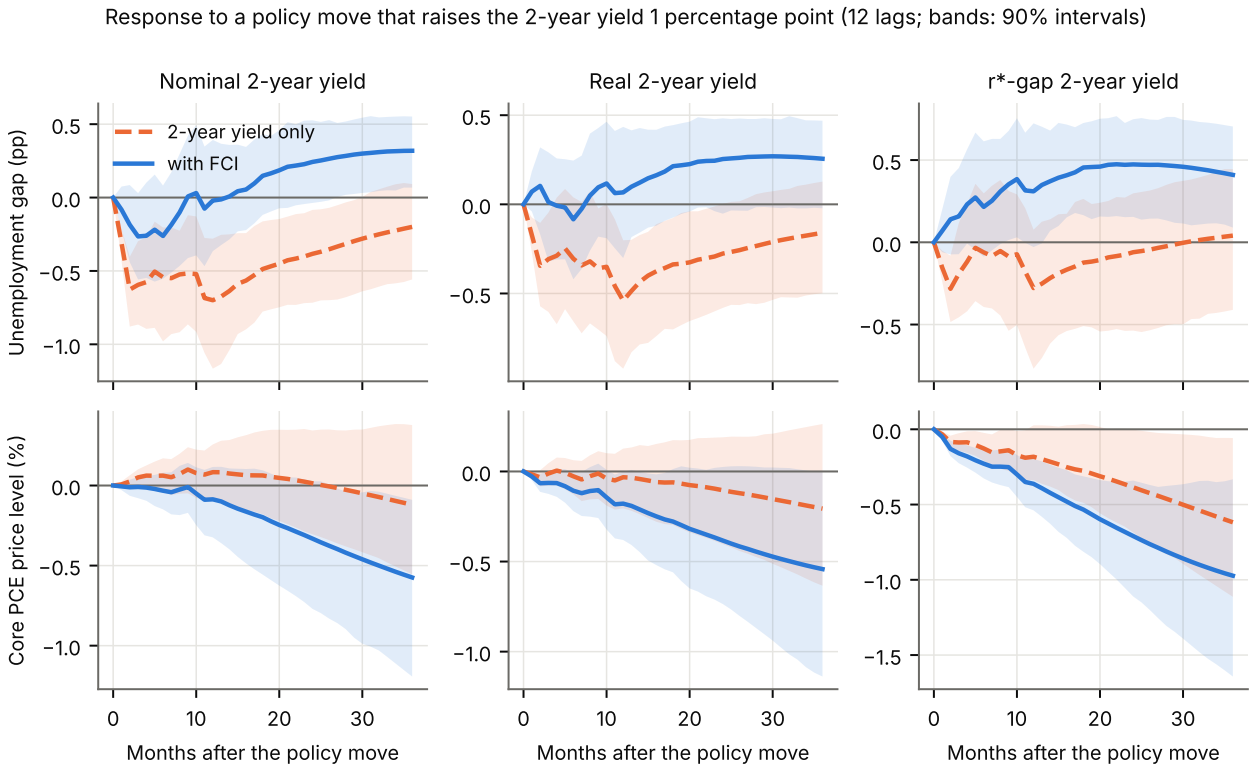


Figure 1: Baseline: the three steps, without and with the FCI

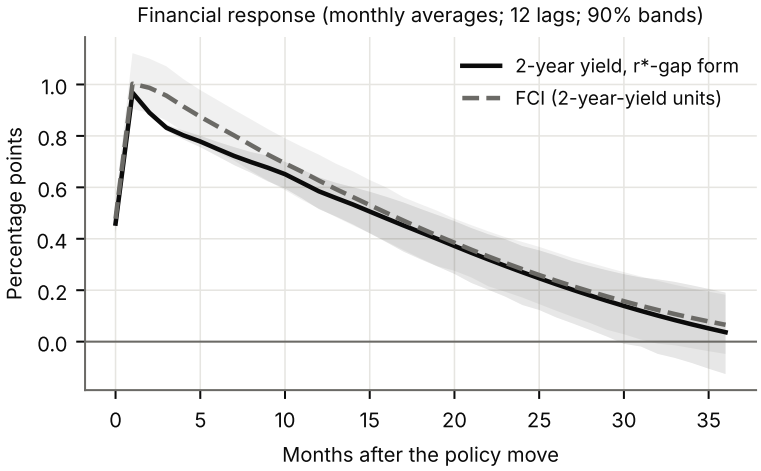


Figure 2: The financial response behind the baseline

Table 2: Market-index weights from policy news

Component	Response to policy news	Weight	Share of a policy move
10-year Treasury yield (pp)	+0.53 (0.06)	+0.609	0.33
Equity valuation, 100 ln CAPE (%)	−7.78 (0.93)	−0.051	0.40
Real broad dollar, 100 ln (%)	+4.22 (0.53)	+0.065	0.28

Response of each 30-minute announcement-window change to the policy-news factor (scaled so the 2-year yield moves 1 pp), with robust standard errors; weights from the one-factor GLS of Note A; share = weight × response (the three shares sum to 1).

Table 3: Implied trade-off: Phillips multiplier κ_H (12 lags)

Case	FCI	$H = 24$ months	$H = 36$ months
Nominal	none	−0.01 [−0.30, 0.47]	0.09 [−0.22, 0.88]
	nominal	unbounded	unbounded
Real	none	0.15 [−0.25, 1.03]	0.23 [−0.23, 1.57]
	real	unbounded	unbounded
r*-gap	none	unbounded	unbounded
	r*-gap	−1.04 [−2.45, −0.46]	−0.86 [−2.24, −0.33]
Start 1973	r*-gap	unbounded	unbounded
2020 dropped	r*-gap	−0.99 [−1.95, −0.31]	−0.83 [−2.22, −0.21]
2020 dropped, start 1973	r*-gap	−1.47 [−2.50, −0.89]	−1.59 [−2.84, −0.90]
Ends 2019	r*-gap	−0.24 [−1.27, 0.22]	unbounded
Ends 2019, start 1973	r*-gap	−1.03 [−1.87, −0.49]	−1.25 [−2.41, −0.68]
r*-gap minus nominal, with FCI		[−1.20, 3.96]	[−1.20, 5.61]

κ_H = core PCE price level at month H (percent) per percentage-point year of cumulative unemployment-gap response (Barnichon and Mesters 2021); the sacrifice ratio is $-1/\kappa_H$. Brackets: 90% confidence set (Anderson–Rubin type; unbounded when it reaches the edge of the grid, −6 to 3). Last row: paired difference over common announcement draws.

2 Robustness

Table 4: Robustness 1: sample (r*-gap 2-year yield and FCI) (12 lags)

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Start 1973	none	12	633	−0.05	0.00	0.00	−0.23*	−0.50*	−0.78*
				[−0.22, 0.13]	[−0.19, 0.17]	[−0.19, 0.24]	[−0.36, −0.11]	[−0.79, −0.24]	[−1.25, −0.31]
	r*-gap	12	633	0.18	0.15	0.14	−0.27*	−0.62*	−0.94*
				[−0.04, 0.37]	[−0.01, 0.31]	[−0.04, 0.32]	[−0.44, −0.12]	[−0.94, −0.29]	[−1.46, −0.42]
2020 dropped	none	12	382	−0.08	−0.15	−0.16	−0.18*	−0.31	−0.44
				[−0.26, 0.14]	[−0.46, 0.22]	[−0.57, 0.31]	[−0.35, −0.02]	[−0.66, 0.07]	[−0.97, 0.12]
	r*-gap	12	382	0.35*	0.41*	0.37*	−0.33*	−0.61*	−0.84*
				[0.15, 0.55]	[0.15, 0.70]	[0.03, 0.73]	[−0.51, −0.12]	[−1.02, −0.15]	[−1.51, −0.15]
2020 dropped, start 1973	none	12	623	0.18*	0.14	0.04	−0.29*	−0.64*	−0.96*
				[0.07, 0.29]	[0.00, 0.30]	[−0.14, 0.22]	[−0.41, −0.16]	[−0.89, −0.37]	[−1.35, −0.58]
	r*-gap	12	623	0.32*	0.24*	0.16	−0.32*	−0.74*	−1.11*
				[0.20, 0.43]	[0.10, 0.39]	[0.00, 0.32]	[−0.46, −0.16]	[−1.05, −0.41]	[−1.69, −0.56]
Ends 2019	none	12	311	−0.14	−0.19	−0.16	0.10	0.10	0.07
				[−0.34, 0.08]	[−0.55, 0.17]	[−0.66, 0.29]	[−0.02, 0.22]	[−0.07, 0.28]	[−0.19, 0.28]
	r*-gap	12	311	0.30*	0.30	0.23	−0.05	−0.12	−0.21
				[0.12, 0.49]	[−0.04, 0.59]	[−0.21, 0.61]	[−0.24, 0.07]	[−0.42, 0.07]	[−0.65, 0.07]
Ends 2019, start 1973	none	12	552	0.20*	0.18*	0.07	−0.22*	−0.51*	−0.83*
				[0.07, 0.35]	[0.01, 0.39]	[−0.14, 0.35]	[−0.35, −0.07]	[−0.82, −0.16]	[−1.36, −0.25]
	r*-gap	12	552	0.33*	0.25*	0.15	−0.23*	−0.54*	−0.90*
				[0.21, 0.47]	[0.10, 0.44]	[−0.03, 0.36]	[−0.41, −0.07]	[−0.94, −0.21]	[−1.53, −0.41]

Start 1973: monthly equations from 1973, with monthly series in place of daily data before 1990 (lib/model.py). 2020 dropped: March to December 2020 removed from the monthly and daily equations (2020 values still enter as lags). Ends 2019: all equations, monthly and daily. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 5: Robustness 2: the r* adjustment and trends (12 lags)

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36
Kiley r*	none	12	392	−0.47	−0.26	−0.17	−0.09	−0.11	−0.
				[−0.91, 0.01]	[−0.59, 0.17]	[−0.54, 0.26]	[−0.26, 0.12]	[−0.49, 0.24]	[−0.67, 0.
	r*-gap (Kiley)	12	392	0.44*	0.42*	0.14	−0.24*	−0.22	−0.
				[0.05, 0.78]	[0.10, 0.79]	[−0.17, 0.48]	[−0.41, −0.05]	[−0.66, 0.19]	[−0.74, 0.
Kiley r*, start 1973	none	12	633	−0.11	−0.05	−0.02	−0.18*	−0.35*	−0.5
				[−0.25, 0.03]	[−0.21, 0.12]	[−0.18, 0.14]	[−0.31, −0.06]	[−0.61, −0.11]	[−0.90, −0.
	r*-gap (Kiley)	12	633	0.05	0.04	0.03	−0.20*	−0.37*	−0.4
				[−0.08, 0.25]	[−0.08, 0.18]	[−0.08, 0.17]	[−0.34, −0.06]	[−0.66, −0.12]	[−0.88, −0.
Real, spline trend	none	12	392	−0.60*	−0.27	−0.10	−0.02	−0.18	−0.
				[−0.94, −0.27]	[−0.63, 0.06]	[−0.49, 0.20]	[−0.19, 0.18]	[−0.52, 0.18]	[−0.91, 0.
	real	12	392	0.27	0.51*	0.42*	−0.18	−0.41	−0.
				[−0.20, 0.70]	[0.00, 1.02]	[0.04, 0.80]	[−0.39, 0.00]	[−0.94, 0.12]	[−1.23, 0.
Real, spline trend, start 1973	none	12	633	−0.10	0.09	0.12	−0.10	−0.25*	−0.3
				[−0.31, 0.06]	[−0.22, 0.31]	[−0.17, 0.34]	[−0.19, 0.00]	[−0.44, −0.02]	[−0.64, −0.
	real	12	633	0.23*	0.27*	0.19*	−0.18*	−0.42*	−0.5
				[0.03, 0.42]	[0.06, 0.48]	[0.02, 0.36]	[−0.30, −0.03]	[−0.66, −0.16]	[−0.89, −0.

Kiley r*: 2-year minus expected inflation minus Kiley's r*, with Note A's index built with the same r*. Real, spline trend: real 2-year and real FCI with a natural cubic spline trend (knots at the sample ends and at NBER business-cycle peaks inside the sample) in every monthly equation, in place of any r* measure. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 6: Robustness 3: excess bond premium (r*-gap) (12 lags)

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
EBP added	none	12	392	−0.09	0.06	0.15	−0.23*	−0.45*	−0.69*
				[−0.47, 0.30]	[−0.37, 0.50]	[−0.30, 0.60]	[−0.39, −0.08]	[−0.82, −0.14]	[−1.23, −0.22]
	r*-gap	12	392	0.23	0.46*	0.36	−0.33*	−0.67*	−0.94*
				[−0.23, 0.68]	[0.00, 0.85]	[−0.11, 0.82]	[−0.54, −0.08]	[−1.14, −0.21]	[−1.71, −0.22]
EBP added, start 1973	none	12	633	−0.03	−0.01	0.01	−0.23*	−0.50*	−0.78*
				[−0.17, 0.15]	[−0.19, 0.16]	[−0.19, 0.18]	[−0.38, −0.11]	[−0.83, −0.23]	[−1.32, −0.28]
	r*-gap	12	633	0.07	0.04	0.04	−0.24*	−0.56*	−0.86*
				[−0.11, 0.25]	[−0.17, 0.24]	[−0.16, 0.24]	[−0.40, −0.08]	[−0.89, −0.22]	[−1.37, −0.37]

The excess bond premium is a third monthly variable, treated like unemployment and inflation (u, inflation, EBP system); responses shown for u and the price level. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

A Results for 3, 6 and 12 lags

Table 7: Baseline: 1994 to the present, all data in, no trend

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Nominal	none	3	392	−0.53*	−0.37*	−0.26*	0.10	0.07	0.04
				[−0.72, −0.27]	[−0.62, −0.07]	[−0.53, −0.01]	[−0.07, 0.28]	[−0.20, 0.35]	[−0.30, 0.40]
		6	392	−0.48*	−0.34*	−0.22	0.09	0.03	−0.08
				[−0.69, −0.20]	[−0.60, −0.05]	[−0.47, 0.13]	[−0.11, 0.28]	[−0.33, 0.36]	[−0.55, 0.37]
		12	392	−0.70*	−0.38*	−0.20	0.08	0.01	−0.12
				[−1.17, −0.28]	[−0.72, −0.08]	[−0.56, 0.09]	[−0.10, 0.26]	[−0.30, 0.34]	[−0.55, 0.38]
	nominal	3	392	0.03	0.18	0.18	−0.06	−0.21	−0.31
				[−0.21, 0.28]	[−0.09, 0.38]	[0.00, 0.31]	[−0.24, 0.14]	[−0.48, 0.12]	[−0.67, 0.15]
		6	392	0.03	0.24	0.27*	−0.08	−0.29	−0.48
				[−0.24, 0.21]	[−0.03, 0.40]	[0.05, 0.44]	[−0.30, 0.11]	[−0.61, 0.06]	[−1.01, 0.06]
		12	392	−0.02	0.24	0.32*	−0.09	−0.33	−0.57*
				[−0.37, 0.39]	[−0.04, 0.53]	[0.07, 0.55]	[−0.32, 0.12]	[−0.77, 0.03]	[−1.19, −0.09]
Real	none	3	392	−0.32*	−0.30*	−0.25	−0.03	−0.05	−0.06
				[−0.53, −0.09]	[−0.55, −0.01]	[−0.49, 0.03]	[−0.18, 0.10]	[−0.28, 0.16]	[−0.36, 0.21]
		6	392	−0.31*	−0.26*	−0.19	0.00	−0.06	−0.16
				[−0.54, −0.11]	[−0.52, −0.02]	[−0.46, 0.07]	[−0.16, 0.18]	[−0.30, 0.27]	[−0.52, 0.27]
		12	392	−0.54*	−0.28	−0.16	−0.03	−0.10	−0.21
				[−0.92, −0.20]	[−0.58, 0.00]	[−0.50, 0.13]	[−0.20, 0.13]	[−0.40, 0.19]	[−0.63, 0.26]
	real	3	392	0.11	0.11	0.08	−0.14	−0.24	−0.30
				[−0.18, 0.40]	[−0.17, 0.35]	[−0.12, 0.22]	[−0.33, 0.06]	[−0.53, 0.08]	[−0.67, 0.11]
		6	392	0.11	0.22	0.20*	−0.14	−0.33	−0.50
				[−0.08, 0.38]	[−0.01, 0.54]	[0.02, 0.46]	[−0.31, 0.06]	[−0.63, 0.03]	[−0.92, 0.07]
		12	392	0.07	0.25	0.26	−0.18*	−0.38	−0.54*
				[−0.30, 0.43]	[−0.05, 0.48]	[−0.02, 0.47]	[−0.38, 0.00]	[−0.74, 0.03]	[−1.14, 0.00]
r*-gap	none	3	392	−0.17	−0.16	−0.13	−0.09	−0.11	−0.13
				[−0.48, 0.06]	[−0.53, 0.13]	[−0.51, 0.15]	[−0.25, 0.07]	[−0.39, 0.17]	[−0.49, 0.26]
		6	392	−0.11	−0.08	−0.03	−0.12	−0.28	−0.44
				[−0.38, 0.13]	[−0.36, 0.20]	[−0.38, 0.26]	[−0.28, 0.06]	[−0.62, 0.08]	[−0.95, 0.09]
		12	392	−0.28	−0.06	0.04	−0.18	−0.39*	−0.62*
				[−0.77, 0.08]	[−0.54, 0.30]	[−0.41, 0.45]	[−0.34, 0.03]	[−0.68, 0.00]	[−1.11, −0.06]
	r*-gap	3	392	0.25	0.17	0.06	−0.17	−0.21	−0.22
				[−0.03, 0.50]	[−0.12, 0.43]	[−0.16, 0.24]	[−0.40, 0.07]	[−0.57, 0.23]	[−0.67, 0.36]
		6	392	0.31*	0.34*	0.25*	−0.27*	−0.53*	−0.73*
				[0.07, 0.63]	[0.10, 0.68]	[0.04, 0.53]	[−0.48, −0.06]	[−0.97, −0.08]	[−1.42, −0.09]
		12	392	0.31	0.47*	0.41*	−0.36*	−0.71*	−0.97*
				[−0.03, 0.72]	[0.15, 0.75]	[0.09, 0.70]	[−0.54, −0.20]	[−1.11, −0.34]	[−1.64, −0.33]

Nominal, real and r*-gap 2-year yield; the FCI is the matching index of Note A in each case. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 8: Robustness 1a: monthly equations start in 1973

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	642	0.01	0.05	0.05	−0.09	−0.15	−0.19
				[−0.10, 0.20]	[−0.10, 0.26]	[−0.10, 0.27]	[−0.23, 0.06]	[−0.48, 0.22]	[−0.71, 0.38]
		6	639	0.00	0.03	0.03	−0.12	−0.31	−0.52
				[−0.13, 0.12]	[−0.14, 0.18]	[−0.15, 0.17]	[−0.24, 0.02]	[−0.60, 0.02]	[−1.02, 0.01]
		12	633	−0.05	0.00	0.00	−0.23*	−0.50*	−0.78*
				[−0.22, 0.13]	[−0.19, 0.17]	[−0.19, 0.24]	[−0.36, −0.11]	[−0.79, −0.24]	[−1.25, −0.31]
	r*-gap	3	642	0.07	0.03	0.00	−0.08	−0.15	−0.23
				[−0.05, 0.18]	[−0.13, 0.17]	[−0.16, 0.16]	[−0.23, 0.07]	[−0.51, 0.18]	[−0.75, 0.28]
		6	639	0.10	0.08	0.07	−0.14	−0.35*	−0.59*
				[−0.06, 0.28]	[−0.11, 0.28]	[−0.13, 0.27]	[−0.29, 0.02]	[−0.67, −0.01]	[−1.11, −0.14]
		12	633	0.18	0.15	0.14	−0.27*	−0.62*	−0.94*
				[−0.04, 0.37]	[−0.01, 0.31]	[−0.04, 0.32]	[−0.44, −0.12]	[−0.94, −0.29]	[−1.46, −0.42]

r*-gap only. Pre-1990 months use monthly series in place of the daily data (see lib/model.py); shocks are the 252 announcements of 1996 on. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 9: Robustness 1b: 2020 dropped from all equations, start 1994

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	382	0.08	0.16	0.20	−0.15	−0.22	−0.28
				[−0.08, 0.26]	[−0.09, 0.42]	[−0.10, 0.50]	[−0.33, 0.03]	[−0.59, 0.15]	[−0.79, 0.23]
		6	382	0.03	0.04	0.05	−0.12	−0.26	−0.39
				[−0.17, 0.19]	[−0.28, 0.31]	[−0.36, 0.39]	[−0.30, 0.04]	[−0.56, 0.11]	[−0.88, 0.19]
		12	382	−0.08	−0.15	−0.16	−0.18*	−0.31	−0.44
				[−0.26, 0.14]	[−0.46, 0.22]	[−0.57, 0.31]	[−0.35, −0.02]	[−0.66, 0.07]	[−0.97, 0.12]
	r*-gap	3	382	0.30*	0.38*	0.34*	−0.18	−0.28	−0.37
				[0.13, 0.47]	[0.13, 0.61]	[0.05, 0.59]	[−0.35, 0.02]	[−0.67, 0.08]	[−0.82, 0.19]
		6	382	0.32*	0.39*	0.38*	−0.25*	−0.48*	−0.68*
				[0.15, 0.49]	[0.11, 0.70]	[0.05, 0.77]	[−0.45, −0.04]	[−0.92, −0.07]	[−1.36, −0.07]
		12	382	0.35*	0.41*	0.37*	−0.33*	−0.61*	−0.84*
				[0.15, 0.55]	[0.15, 0.70]	[0.03, 0.73]	[−0.51, −0.12]	[−1.02, −0.15]	[−1.51, −0.15]

r*-gap only. March to December 2020 dropped from the monthly and the daily equations. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 10: Robustness 1c: 2020 dropped from all equations, start 1973

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	632	0.06	0.11	0.13	−0.11	−0.20	−0.27
				[−0.02, 0.14]	[−0.04, 0.24]	[−0.04, 0.30]	[−0.26, 0.03]	[−0.53, 0.13]	[−0.80, 0.22]
		6	629	0.11*	0.10	0.04	−0.17*	−0.39*	−0.62*
				[0.01, 0.22]	[−0.06, 0.24]	[−0.15, 0.22]	[−0.30, −0.03]	[−0.70, −0.05]	[−1.14, −0.12]
		12	623	0.18*	0.14	0.04	−0.29*	−0.64*	−0.96*
				[0.07, 0.29]	[0.00, 0.30]	[−0.14, 0.22]	[−0.41, −0.16]	[−0.89, −0.37]	[−1.35, −0.58]
	r*-gap	3	632	0.12*	0.10	0.08	−0.09	−0.24*	−0.44*
				[0.03, 0.22]	[−0.04, 0.25]	[−0.08, 0.24]	[−0.25, 0.00]	[−0.57, −0.01]	[−0.90, −0.05]
		6	629	0.27*	0.16*	0.07	−0.20*	−0.48*	−0.77*
				[0.16, 0.39]	[0.00, 0.32]	[−0.12, 0.26]	[−0.39, −0.07]	[−0.90, −0.18]	[−1.37, −0.30]
		12	623	0.32*	0.24*	0.16	−0.32*	−0.74*	−1.11*
				[0.20, 0.43]	[0.10, 0.39]	[0.00, 0.32]	[−0.46, −0.16]	[−1.05, −0.41]	[−1.69, −0.56]

r*-gap only. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 11: Robustness 1d: sample ends in December 2019, start 1994

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	311	−0.09	−0.13	−0.15	0.02	0.08	0.13
				[−0.26, 0.12]	[−0.43, 0.18]	[−0.50, 0.25]	[−0.05, 0.11]	[−0.04, 0.22]	[−0.04, 0.30]
		6	311	−0.09	−0.14	−0.15	0.04	0.08	0.10
				[−0.33, 0.13]	[−0.51, 0.23]	[−0.59, 0.36]	[−0.05, 0.14]	[−0.12, 0.24]	[−0.19, 0.33]
		12	311	−0.14	−0.19	−0.16	0.10	0.10	0.07
				[−0.34, 0.08]	[−0.55, 0.17]	[−0.66, 0.29]	[−0.02, 0.22]	[−0.07, 0.28]	[−0.19, 0.28]
	r*-gap	3	311	0.20*	0.21	0.17	−0.04	0.00	0.03
				[0.00, 0.44]	[−0.10, 0.54]	[−0.17, 0.57]	[−0.14, 0.07]	[−0.17, 0.20]	[−0.19, 0.28]
		6	311	0.23*	0.21	0.17	−0.07	−0.04	−0.04
				[0.01, 0.48]	[−0.14, 0.66]	[−0.27, 0.74]	[−0.19, 0.05]	[−0.23, 0.17]	[−0.26, 0.23]
		12	311	0.30*	0.30	0.23	−0.05	−0.12	−0.21
				[0.12, 0.49]	[−0.04, 0.59]	[−0.21, 0.61]	[−0.24, 0.07]	[−0.42, 0.07]	[−0.65, 0.07]

r*-gap only. All equations, monthly and daily. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 12: Robustness 1e: sample ends in December 2019, start 1973

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	561	0.03	0.05	0.06	−0.05	−0.06	−0.07
				[−0.06, 0.13]	[−0.09, 0.21]	[−0.10, 0.24]	[−0.19, 0.12]	[−0.45, 0.37]	[−0.67, 0.60]
		6	558	0.11*	0.09	0.04	−0.10	−0.27	−0.46
				[0.01, 0.24]	[−0.05, 0.28]	[−0.12, 0.23]	[−0.24, 0.05]	[−0.58, 0.07]	[−1.06, 0.10]
		12	552	0.20*	0.18*	0.07	−0.22*	−0.51*	−0.83*
				[0.07, 0.35]	[0.01, 0.39]	[−0.14, 0.35]	[−0.35, −0.07]	[−0.82, −0.16]	[−1.36, −0.25]
	r*-gap	3	561	0.08*	0.04	0.00	−0.01	−0.06	−0.19
				[0.00, 0.18]	[−0.11, 0.19]	[−0.18, 0.19]	[−0.14, 0.14]	[−0.33, 0.24]	[−0.60, 0.25]
		6	558	0.26*	0.13	0.02	−0.13	−0.34	−0.59*
				[0.15, 0.39]	[−0.04, 0.28]	[−0.15, 0.20]	[−0.28, 0.01]	[−0.67, 0.02]	[−1.12, −0.06]
		12	552	0.33*	0.25*	0.15	−0.23*	−0.54*	−0.90*
				[0.21, 0.47]	[0.10, 0.44]	[−0.03, 0.36]	[−0.41, −0.07]	[−0.94, −0.21]	[−1.53, −0.41]

r*-gap only. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 13: Robustness 2a: Kiley r*, start 1994

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap (Kiley)	none	3	392	−0.31*	−0.32*	−0.27	−0.05	−0.03	−0.02
				[−0.50, −0.05]	[−0.55, −0.03]	[−0.50, 0.00]	[−0.23, 0.10]	[−0.28, 0.21]	[−0.34, 0.32]
		6	392	−0.27*	−0.26*	−0.20	−0.05	−0.10	−0.17
				[−0.64, −0.06]	[−0.58, 0.00]	[−0.54, 0.07]	[−0.21, 0.14]	[−0.41, 0.22]	[−0.64, 0.29]
		12	392	−0.47	−0.26	−0.17	−0.09	−0.11	−0.14
				[−0.91, 0.01]	[−0.59, 0.17]	[−0.54, 0.26]	[−0.26, 0.12]	[−0.49, 0.24]	[−0.67, 0.35]
	r*-gap (Kiley)	3	392	0.24	0.08	−0.01	−0.09	−0.07	−0.06
				[−0.08, 0.63]	[−0.16, 0.33]	[−0.19, 0.13]	[−0.29, 0.16]	[−0.40, 0.35]	[−0.45, 0.43]
		6	392	0.40*	0.28	0.11	−0.18	−0.25	−0.30
				[0.10, 0.81]	[−0.01, 0.72]	[−0.13, 0.41]	[−0.43, 0.03]	[−0.75, 0.19]	[−1.06, 0.40]
		12	392	0.44*	0.42*	0.14	−0.24*	−0.22	−0.07
				[0.05, 0.78]	[0.10, 0.79]	[−0.17, 0.48]	[−0.41, −0.05]	[−0.66, 0.19]	[−0.74, 0.50]

2-year minus expected inflation minus Kiley's r*; the FCI is Note A's index built with Kiley's r* (same adjustment in rate and FCI). Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 14: Robustness 2b: Kiley r*, start 1973

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap (Kiley)	none	3	642	−0.02	0.00	0.01	−0.04	−0.02	0.01
				[−0.16, 0.09]	[−0.15, 0.18]	[−0.13, 0.18]	[−0.17, 0.10]	[−0.33, 0.29]	[−0.46, 0.52]
		6	639	−0.04	−0.01	0.00	−0.07	−0.17	−0.28
				[−0.15, 0.07]	[−0.16, 0.16]	[−0.16, 0.15]	[−0.18, 0.04]	[−0.42, 0.08]	[−0.70, 0.15]
		12	633	−0.11	−0.05	−0.02	−0.18*	−0.35*	−0.50*
				[−0.25, 0.03]	[−0.21, 0.12]	[−0.18, 0.14]	[−0.31, −0.06]	[−0.61, −0.11]	[−0.90, −0.10]
	r*-gap (Kiley)	3	642	−0.02	−0.05	−0.05	−0.04	−0.06	−0.07
				[−0.11, 0.11]	[−0.15, 0.08]	[−0.15, 0.05]	[−0.17, 0.08]	[−0.29, 0.15]	[−0.38, 0.23]
		6	639	−0.01	−0.01	0.00	−0.08	−0.18	−0.27
				[−0.12, 0.12]	[−0.14, 0.13]	[−0.13, 0.14]	[−0.20, 0.04]	[−0.42, 0.06]	[−0.63, 0.08]
		12	633	0.05	0.04	0.03	−0.20*	−0.37*	−0.48*
				[−0.08, 0.25]	[−0.08, 0.18]	[−0.08, 0.17]	[−0.34, −0.06]	[−0.66, −0.12]	[−0.88, −0.15]

As 2a. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 15: Robustness 2c: real 2-year and real FCI with a cubic spline trend, start 1994

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Real	none	3	392	−0.42*	−0.35*	−0.28*	−0.10	−0.20	−0.29
				[−0.74, −0.11]	[−0.72, −0.05]	[−0.65, −0.02]	[−0.29, 0.06]	[−0.53, 0.07]	[−0.69, 0.09]
		6	392	−0.42*	−0.29	−0.17	−0.06	−0.29	−0.51*
				[−0.74, −0.10]	[−0.70, 0.08]	[−0.63, 0.19]	[−0.22, 0.13]	[−0.58, 0.07]	[−0.91, −0.03]
		12	392	−0.60*	−0.27	−0.10	−0.02	−0.18	−0.42
				[−0.94, −0.27]	[−0.63, 0.06]	[−0.49, 0.20]	[−0.19, 0.18]	[−0.52, 0.18]	[−0.91, 0.11]
	real	3	392	0.04	0.06	0.05	−0.17	−0.31	−0.39
				[−0.28, 0.45]	[−0.23, 0.39]	[−0.15, 0.19]	[−0.36, 0.02]	[−0.64, 0.05]	[−0.77, 0.07]
		6	392	0.17	0.32	0.28*	−0.22	−0.55	−0.76
				[−0.21, 0.53]	[−0.08, 0.63]	[0.00, 0.50]	[−0.45, 0.04]	[−1.02, 0.04]	[−1.39, 0.02]
		12	392	0.27	0.51*	0.42*	−0.18	−0.41	−0.53
				[−0.20, 0.70]	[0.00, 1.02]	[0.04, 0.80]	[−0.39, 0.00]	[−0.94, 0.12]	[−1.23, 0.27]

No r* measure: the trend is a natural cubic spline in time (knots at the sample ends and at the NBER business-cycle peaks inside the sample), estimated in every monthly equation. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 16: Robustness 2d: real 2-year and real FCI with a cubic spline trend, start 1973

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Real	none	3	642	0.01	0.04	0.04	−0.11*	−0.22*	−0.29*
				[−0.17, 0.19]	[−0.18, 0.24]	[−0.15, 0.19]	[−0.23, −0.01]	[−0.44, −0.03]	[−0.60, −0.05]
		6	639	0.02	0.07	0.06	−0.11*	−0.32*	−0.48*
				[−0.14, 0.14]	[−0.14, 0.24]	[−0.13, 0.20]	[−0.23, −0.01]	[−0.55, −0.04]	[−0.81, −0.13]
		12	633	−0.10	0.09	0.12	−0.10	−0.25*	−0.39*
				[−0.31, 0.06]	[−0.22, 0.31]	[−0.17, 0.34]	[−0.19, 0.00]	[−0.44, −0.02]	[−0.64, −0.06]
	real	3	642	0.26*	0.23*	0.15*	−0.18*	−0.33*	−0.41*
				[0.06, 0.43]	[0.06, 0.37]	[0.04, 0.24]	[−0.28, −0.05]	[−0.52, −0.07]	[−0.65, −0.08]
		6	639	0.27*	0.29*	0.19*	−0.18*	−0.46*	−0.65*
				[0.06, 0.48]	[0.08, 0.49]	[0.06, 0.34]	[−0.32, −0.05]	[−0.73, −0.16]	[−1.00, −0.25]
		12	633	0.23*	0.27*	0.19*	−0.18*	−0.42*	−0.57*
				[0.03, 0.42]	[0.06, 0.48]	[0.02, 0.36]	[−0.30, −0.03]	[−0.66, −0.16]	[−0.89, −0.19]

As 2c; knots at the NBER peaks of 1973, 1980, 1981, 1990, 2001, 2007 and 2020. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 17: Robustness 3a: excess bond premium as a third monthly variable, start 1994

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	392	−0.10	−0.13	−0.12	−0.12	−0.15	−0.16
				[−0.40, 0.18]	[−0.48, 0.17]	[−0.46, 0.16]	[−0.31, 0.04]	[−0.50, 0.14]	[−0.62, 0.23]
		6	392	0.00	−0.07	−0.09	−0.15	−0.29	−0.44
				[−0.34, 0.31]	[−0.44, 0.32]	[−0.52, 0.28]	[−0.33, 0.02]	[−0.69, 0.01]	[−1.00, 0.04]
		12	392	−0.09	0.06	0.15	−0.23*	−0.45*	−0.69*
				[−0.47, 0.30]	[−0.37, 0.50]	[−0.30, 0.60]	[−0.39, −0.08]	[−0.82, −0.14]	[−1.23, −0.22]
	r*-gap	3	392	0.19	0.06	−0.10	−0.15	−0.16	−0.14
				[−0.10, 0.50]	[−0.29, 0.50]	[−0.46, 0.37]	[−0.35, 0.01]	[−0.53, 0.13]	[−0.63, 0.27]
		6	392	0.28	0.20	0.03	−0.23*	−0.43*	−0.61*
				[−0.01, 0.57]	[−0.17, 0.59]	[−0.34, 0.49]	[−0.45, −0.04]	[−0.87, −0.05]	[−1.22, −0.02]
		12	392	0.23	0.46*	0.36	−0.33*	−0.67*	−0.94*
				[−0.23, 0.68]	[0.00, 0.85]	[−0.11, 0.82]	[−0.54, −0.08]	[−1.14, −0.21]	[−1.71, −0.22]

r*-gap. Monthly system u, inflation, EBP (EBP treated like u and inflation; responses shown for u and the price level). Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table 18: Robustness 3b: excess bond premium as a third monthly variable, start 1973

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	642	0.01	0.04	0.04	−0.09	−0.15	−0.19
				[−0.10, 0.15]	[−0.10, 0.21]	[−0.09, 0.19]	[−0.25, 0.06]	[−0.52, 0.23]	[−0.72, 0.42]
		6	639	0.01	0.03	0.01	−0.12*	−0.31*	−0.51*
				[−0.10, 0.14]	[−0.14, 0.19]	[−0.19, 0.16]	[−0.26, −0.02]	[−0.61, −0.04]	[−1.06, −0.09]
		12	633	−0.03	−0.01	0.01	−0.23*	−0.50*	−0.78*
				[−0.17, 0.15]	[−0.19, 0.16]	[−0.19, 0.18]	[−0.38, −0.11]	[−0.83, −0.23]	[−1.32, −0.28]
	r*-gap	3	642	0.03	0.02	0.00	−0.09	−0.19	−0.28
				[−0.09, 0.14]	[−0.13, 0.16]	[−0.16, 0.14]	[−0.22, 0.08]	[−0.51, 0.18]	[−0.75, 0.29]
		6	639	0.04	0.03	0.01	−0.12	−0.31	−0.55
				[−0.09, 0.19]	[−0.15, 0.18]	[−0.18, 0.15]	[−0.26, 0.03]	[−0.62, 0.02]	[−1.04, 0.04]
		12	633	0.07	0.04	0.04	−0.24*	−0.56*	−0.86*
				[−0.11, 0.25]	[−0.17, 0.24]	[−0.16, 0.24]	[−0.40, −0.08]	[−0.89, −0.22]	[−1.37, −0.37]

As 3a. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.